SAHIL KUMAR SHARMA

Software Engineer | Backend Development, AWS & Applied AI

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PROFESSIONAL SUMMARY

Software engineer who turns business requirements into backend services, cloud workflows and AI features. Experience spans production APIs, payment integrations, conversational systems and performance optimization. Independent work includes deployed applications with retrieval, agent orchestration, model inference and automated evaluation, supported by persistent data and tested APIs.

TECHNICAL SKILLS

- **Languages**: Python, Java, SQL, JavaScript, TypeScript
- **Backend**: FastAPI, Flask, Node.js, Express.js, Spring Boot, REST APIs, OpenAPI/Swagger, Pydantic
- **Databases & Persistence**: PostgreSQL, DynamoDB, SQLite, SQLAlchemy, Prisma
- **AWS**: Lambda, API Gateway, S3, IAM, Amazon Bedrock, Amazon Connect, Amazon Lex
- **Generative AI**: LangChain, CrewAI, Google Gemini, Retrieval-Augmented Generation (RAG), ChromaDB, Prompt Engineering, Tool Calling, Structured Outputs, LLM Evaluation
- **Data & Machine Learning**: NumPy, Pandas, scikit-learn, TF-IDF, Ridge Regression, Gradient Boosting, ONNX Runtime, InsightFace/ArcFace
- **Testing & Delivery**: Pytest, Vitest, Playwright, Git, GitHub Actions, CI/CD, Docker
- **Frontend & Security**: React, Next.js, JWT Authentication, Role-Based Access Control

EXPERIENCE

Tata Consultancy Services (TCS) | *Cloud Application Engineer* December 2025 - Present

Python, AWS Lambda, Amazon Bedrock, DynamoDB, S3, API Gateway, Amazon Connect, Amazon Lex

- Engineered an AI-powered lead-processing pipeline using Python, AWS Lambda, Amazon Bedrock and S3, translating plain-English business requests into executable filtering rules for batches of 10K+ records.
- Implemented generated-rule validation, automated retries and a four-table DynamoDB architecture for workflow state management, integrating real-time Do Not Disturb registry checks and timestamp normalization to Indian Standard Time.
- Optimized 5+ event-driven backend services and API Gateway endpoints, reducing rule-processing latency to under two seconds and manual campaign operations effort by 45%.
- Modernized legacy DTMF-based IVRs using Amazon Connect and Amazon Lex across 15K+ monthly interactions; configured contact flows, queues and routing profiles with Lambda and REST API integrations for customer lookup and dynamic routing.

Fundbridge Consultation Ltd. | *SDE Intern* July 2025 - December 2025

Node.js, Express.js, PostgreSQL, REST APIs

- Developed 13+ RESTful API endpoints for investment and administration workflows, implementing authenticated CRUD operations and PostgreSQL persistence.
- Integrated payment gateway services and investigated production issues affecting transaction processing and application stability, tracing failures across APIs and application logic.
- Delivered full-stack features covering authentication, media management and automated financial calculations, connecting frontend interactions with backend services.

PlayZarro | *Freelance Full-Stack Developer* April 2025 - June 2025

TypeScript, Next.js, React, PostgreSQL, Prisma, Vitest, Playwright

- Developed backend features for a gaming platform using Next.js, PostgreSQL and Prisma, implementing server-side sessions with hashed tokens, bcrypt password hashing and role-based access for player and administration workflows.
- Separated game rules into independently testable engines and built 295 deterministic tests across 10 files covering valid actions, state transitions and chip conservation; added Playwright end-to-end flows and layout checks across five viewport sizes.
- Profiled and resolved rendering bottlenecks, improving poker rendering from 14 to 60 FPS through changes to simulation execution and expensive visual effects.

PROJECTS

FlowMancer - AI Operations & Workflow Platform
Live GitHub

Python, FastAPI, CrewAI, LangChain, Gemini, ChromaDB, PostgreSQL, Docker

- Built lead qualification, email triage and document-processing workflows with schema-validated LLM outputs, supporting single-call processing and four-agent CrewAI pipelines.
- Implemented RAG using document chunking, local embeddings and ChromaDB retrieval; developed an account-scoped tool-calling assistant that searches business records and submits proposed actions for human approval.
- Deployed the containerized API on Render with PostgreSQL on Neon, persisting workflow state and streaming execution progress through Server-Sent Events; added model fallback and token, latency and estimated-cost tracking.
- Added 114 backend tests and a custom evaluation suite covering 30 labelled cases, with GitHub Actions running regression checks and enforcing a 90% evaluation pass threshold.

SkillLens - AI Career Analytics Platform
Live GitHub

Python, Flask, PostgreSQL, SQLAlchemy, scikit-learn, Gemini API, Docker

- Built career analytics over 97,316 Indian job postings and trained salary models on 31,566 salary-labelled records using Ridge regression, gradient boosting and TF-IDF features, evaluating against companies excluded from training.
- Developed Flask APIs, PostgreSQL persistence and versioned database migrations for resume analysis, job matching and administration; implemented SQL-backed analytics, response caching and external-service usage limits.
- Integrated Gemini feedback with structured JSON responses, personal-data redaction and checks against unsupported numerical claims; configured CI for 178 backend tests, 40 frontend tests and Docker builds.

Smart Attendance System - Face Recognition Platform
Live GitHub

Python, FastAPI, NumPy, InsightFace/ArcFace, PostgreSQL, SQLAlchemy, Docker

- Developed FastAPI services for face enrollment, attendance records and analytics, implementing JWT authentication, admin/user authorization and PostgreSQL persistence through SQLAlchemy.
- Integrated ArcFace's 512-dimensional embeddings with NumPy-based vectorized matching and an enrollment-aware embedding cache, incorporating quality checks, duplicate-face detection and multi-frame voting.
- Containerized the backend for Hugging Face Spaces with the frontend on Vercel; persisted enrollment data across redeployments and configured GitHub Actions for Pytest authentication and authorization tests and frontend builds.

SELECTED TRAINING & ACHIEVEMENTS

Amazon ML Summer School | Selected Participant | 2024

- Selected for and attended Amazon ML Summer School 2024, covering eight machine learning topics, including supervised learning, deep neural networks, generative AI, large language models and reinforcement learning.

EY Global Delivery Services | Cyber Security Boot Camp | 2024

- Completed the Cyber Security Boot Camp under the Curriculum Immersion Program and earned recognition as a Top 1% Performer.

CERTIFICATIONS

- **AWS Certified Cloud Practitioner CLF-C02** | 2026
- **AWS Academy Cloud Foundations, Architecting & Security** | 2024

EDUCATION

ITER, Siksha 'O' Anusandhan University — **November 2021 - August 2025**

B.Tech, Computer Science and Engineering — Bhubaneswar, Odisha